GHamine

Systems engineer

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SUMMARY

I build full-stack systems and measure whether their stated invariants hold. Each case study on this site carries the artifact path and the limits behind every number.

SELECTED SYSTEMS

Full-Stack E-Commerce Platform

Next.js, TypeScript, PostgreSQL, Stripe

- 20 concurrent checkouts against stock 5 produced 5 orders, 15 rejections, stock 0 and 5 open reservations; a conditional UPDATE and CHECK (stock >= 0) prevent oversell.
- 50 VUs x 5 iterations on k6: 250 checkouts with 0% errors and p95 2.66 s on a Next.js dev server. The five-minute profile was not run, and the load report states that.

CI/CD pipeline with gates that actually block

GitHub Actions, Docker Buildx, Trivy, cosign, Fastify, PostgreSQL

- Eight PR gates feed one required ci-ok check; a structural check found 9 jobs and 0 unpinned uses. No Actions run exists, so red-PR blocking is not measured.
- A release whose POST /invoices returned 500 failed the smoke gate and rolled back to the previous release; deploy.log records the order with timestamps.

DocQA Assistant

Python, FastAPI, PostgreSQL + pgvector, React

- recall@8 18/24 (75.0%, Wilson 95% [55.1%, 88.0%]) on a 24-answerable eval set; the committed report fails answer rate, citation precision and faithfulness gates and exits 6.
- Every claim needs an in-context citation whose quote is a normalised substring of the cited chunk; tests/test_citations.py fails if an uncited claim survives. Answer quality under a generator model is not measured because no model key exists.

SKILLS

PostgreSQL / Node.js / TypeScript / Next.js / Python / FastAPI / pgvector / GitHub Actions / Docker / Stripe / Astro

VERIFICATION

- npm run verify runs astro check, tsc, the asset and content contracts, the copy and accessibility audits, the JS budget, and a static build.
- Every metric on a case page is read from public/assets/<slug>/manifest.json and links to its committed artifact.
- The site ships no client-side JavaScript and no tracking.

LIMITS

- The e-commerce load run used 250 checkouts on one Windows machine, not the planned production profile.

- The CI repository has no remote: workflow structure is verified, Actions execution is not.
- DocQA has no generator model key; answers use the extractive fallback and three eval gates fail.
- The deployments are local only; no public URL exists for any of the three systems.

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